

SCX Self-Evolution as a Discrete Dynamical System

Version: 2026-06-28 | **Status:** Theoretical development | **Audit:** Pre-verification
Purpose: Formalize the SCX self-evolution loop as a discrete dynamical system, study fixed points, Lyapunov stability, attractor structure, and phase portraits. **Prerequisites:** Document 01 (Symbol System and Problem Setup)

Table of Contents

1. Discrete Dynamical System Formulation
 2. State Space Topology and Metric on S_t
 3. Orbits Under Φ
 4. Attractors and Omega-Limit Sets
 5. Fixed Points
 6. Existence Conditions for Fixed Points
 7. Lyapunov Function Candidate
 8. Monotonicity Analysis
 9. Phase Portrait
 10. Connection to Existing Theorems 1-3
 11. Summary of Proven vs. Conjectured Claims
-

1. Discrete Dynamical System Formulation

1.1 Basic Formulation

Let $z_t = (S_t, M_t, \theta_t) \in \mathcal{Z}$ be the evolution state at time t as defined in Document 01, Section 5. The self-evolution loop is a discrete dynamical system:

$$z_{t+1} = \Phi(z_t), \quad t = 0, 1, 2, \dots$$

with initial condition $z_0 = (S_0, M_0, \theta_0)$ determined by the static SCX initialization.

Definition 16 (Discrete-Time Dynamical System). The tuple $(\mathcal{Z}, \Phi, \mathbb{N}_0)$ constitutes a discrete-time dynamical system where: - $\mathcal{Z} = \mathcal{F} \times \mathcal{M} \times \Theta$ is the state space (Section 5 of Document 01) - $\Phi : \mathcal{Z} \rightarrow \mathcal{Z}$ is the update operator (Section 9 of Document 01) - $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ is the discrete time index

1.2 Decomposition into Component Maps

The operator Φ decomposes into three coupled maps:

$$\Phi(S, M, \theta) = (\Phi_S(S, M, \theta), \Phi_M(S), \Phi_\theta(S, M))$$

where the coupling structure is:

$$\begin{aligned} S_t & \xrightarrow{\Phi_M} M_{t+1} \\ S_t & \xrightarrow{\Phi_S} S_{t+1} \quad (\text{via } M_{t+1}) \end{aligned}$$

$$S_t, M_{t+1}, \theta_{t+1} \leftarrow \Phi_S \rightarrow S_{t+1}$$

This coupling is **feed-forward**: S_t influences M_{t+1} and θ_{t+1} , which in turn influence S_{t+1} . There is no instantaneous feedback loop.

Proposition 3 (Causal Structure). The system's information flow is a directed acyclic graph within each timestep: $S_t \rightarrow M_{t+1} \rightarrow \theta_{t+1}$, and $(S_t, M_{t+1}, \theta_{t+1}) \rightarrow S_{t+1}$. Consequently, the Jacobian of Φ is block-triangular.

Proof. By inspection of the update definitions: Φ_M depends only on S_t (not on M_t or θ_t); Φ_θ depends on M_{t+1} and implicitly on S_t through M_{t+1} ; Φ_S depends on all three. $\square$

2. State Space Topology and Metric on S_t

2.1 Metric Structure

Recall from Document 01 the metric on $\mathcal{F}$:

$$d_{\mathcal{F}}(S, S') = \mathbb{E}_{(x,y) \sim \mathcal{D}} [|S(x, y) - S'(x, y)|]$$

and the product metric on $\mathcal{Z}$:

$$d_{\mathcal{Z}}(z, z') = d_{\mathcal{F}}(S, S') + d_{\mathcal{M}}(M, M') + \|\theta - \theta'\|_2$$

Proposition 4 (Completeness). If $\mathcal{D}$ has full support on $\mathcal{X}_0 \times \mathcal{Y}$, then $(\mathcal{F}, d_{\mathcal{F}})$ is a complete metric space. The product space $(\mathcal{Z}, d_{\mathcal{Z}})$ is complete.

Proof sketch. $L^1(\mathcal{D})$ is complete. The finite set $\mathcal{M}$ with the discrete topology is trivially complete. $\Theta \subseteq \mathbb{R}^{d_\theta}$ is complete if closed. The product of complete metric spaces is complete. $\square$

2.2 Compactness Considerations

For existence of fixed points and attractors, compactness of the state space is desirable.

Proposition 5 (Compactness of $\mathcal{F}$ under Regularization). If we restrict $\mathcal{F}$ to the set of functions: - bounded in L^∞ norm (already $[0, 1]$ -valued) - with uniformly bounded Lipschitz constant $L_S < \infty$ with respect to input distance then $\mathcal{F}$ is compact in $d_{\mathcal{F}}$ by the Arzela-Ascoli theorem (on compact subsets of $\mathcal{X}_0$).

Practical implication. In implementation, neural network gatekeepers satisfy Lipschitz bounds (e.g., via spectral normalization). Theoretical analysis often assumes such regularization to ensure well-posedness.

3. Orbits Under Φ

3.1 Definition

Definition 17 (Orbit). The forward orbit of an initial state $z_0 \in \mathcal{Z}$ under Φ is the sequence:

$$\mathcal{O}^+(z_0) = \{z_0, \Phi(z_0), \Phi^{(2)}(z_0), \Phi^{(3)}(z_0), \dots\}$$

The backward orbit (if Φ is invertible) is defined analogously but is not guaranteed.

3.2 Types of Orbits

Based on the system's behavior, we classify orbits into three types:

Definition 18 (Orbit Classification).

1. **Convergent orbit:** $\lim_{t \rightarrow \infty} d_{\mathcal{Z}}(z_t, z^*) = 0$ for some $z^* \in \mathcal{Z}$.
2. **Periodic orbit:** $\exists T > 0$ such that $z_{t+T} = z_t$ for all sufficiently large t , with no smaller period.
3. **Chaotic orbit:** The orbit is bounded but does not converge and is not periodic; sensitive dependence on initial conditions may occur.

Conjecture 2 (Orbit Classification for SCX). Under Assumptions B1-B3, all orbits of the SCX self-evolution system are convergent. No periodic or chaotic orbits exist.

*Status: **Conjecture.** The monotonicity of M_t (B1) prevents cycling through memory loss, but stochastic gradient updates in Φ_S and Φ_θ could, in principle, produce limit cycles. The conjecture asserts that the Lyapunov function (Section 7) eliminates this possibility.*

3.3 Properties of Orbits

Proposition 6 (Boundedness of Orbits). Under Assumption B2 (bounded memory growth), any orbit $\mathcal{O}^+(z_0)$ is bounded in $\mathcal{Z}$.

Proof. S_t is bounded in $[0, 1]$ by construction, $\|S_t\|_\infty \leq 1$. θ_t is bounded if the loss landscape has bounded sublevel sets (e.g., under L2 regularization).

M_t has bounded size by B2. Hence $d_{\mathcal{Z}}(z_0, z_t)$ is bounded for all t . $\square$

4. Attractors and ω -Limit Sets

4.1 Definition

Definition 19 (ω -Limit Set). For a trajectory $z_t = \Phi^{(t)}(z_0)$, the ω -limit set is:

$$\omega(z_0) = \{z \in \mathcal{Z} : \exists \{t_k\}_{k=1}^{\infty} \text{ with } t_k \rightarrow \infty \text{ and } \lim_{k \rightarrow \infty} d_{\mathcal{Z}}(z_{t_k}, z) = 0\}$$

The α -limit set (for backward orbits) is defined analogously.

Definition 20 (Attractor). A set $A \subseteq \mathcal{Z}$ is an attractor if: 1. A is invariant: $\Phi(A) = A$. 2. There exists a neighborhood U of A such that $\omega(z) \subseteq A$ for all $z \in U$. 3. A is minimal with respect to properties 1 and 2.

The **basin of attraction** of A is:

$$\mathcal{B}(A) = \{z \in \mathcal{Z} : \omega(z) \subseteq A\}$$

4.2 Candidate Attractors for SCX

Proposition 7 (Fixed Points as Attractors). If z^* is an asymptotically stable fixed point (Definition 21), then $\{z^*\}$ is an attractor with basin $\mathcal{B}(z^*)$ containing an open neighborhood of z^* .

The primary candidate attractor for the SCX self-evolution system is the **perfect gatekeeper**:

$$S^*(x, y) = \mathbf{1}\{y = f^*(x)\}$$

i.e., the gatekeeper that perfectly identifies whether the observed label matches the true oracle. However, achieving this fixed point requires: 1. The NEP student converges to f^* (Assumption B3) 2. The memory bank contains sufficient clean samples

Conjecture 3 (Unique Attractor). Under Assumptions B1-B3 and the uniform noise assumption (A4), the SCX self-evolution system has a unique attractor $z^* = (S^*, M^*, \theta^*)$ where $S^*(x, y) = \mathbf{1}\{y = f^*(x)\}$, M^* is the infinite memory bank containing all sufficiently clean samples, and θ^* parameterizes f^* .

*Status: **Conjecture.** The uniqueness depends on the loss landscape of the gatekeeper and NEP being convex, which is not guaranteed for neural network parameterizations. However, if the function spaces are sufficiently expressive and the losses are proper scoring rules, the fixed point is unique in function space.*

5. Fixed Points

5.1 Definition

Definition 21 (Fixed Point). A state $z^* = (S^*, M^*, \theta^*) \in \mathcal{Z}$ is a fixed point of the dynamics if:

$$\Phi(z^*) = z^*$$

Equivalently:

$$\Phi_S(S^*, M^*, \theta^*) = S^*$$

$$\Phi_M(S^*, M^*) = M^*$$

$$\Phi_\theta(S^*, M^*) = \theta^*$$

Definition 22 (Stability of Fixed Points). A fixed point z^* is: - **Lyapunov stable** if $\forall \varepsilon > 0, \exists \delta > 0$ such that $d_{\mathcal{Z}}(z_0, z^*) < \delta$ implies $d_{\mathcal{Z}}(z_t, z^*) < \varepsilon$ for all $t \geq 0$. - **Asymptotically stable** if it is Lyapunov stable and $\exists \delta > 0$ such that $d_{\mathcal{Z}}(z_0, z^*) < \delta$ implies $\lim_{t \rightarrow \infty} d_{\mathcal{Z}}(z_t, z^*) = 0$. - **Unstable** if it is not Lyapunov stable.

5.2 Fixed Point Equations

Fixed point equation for S^* :

$$S^*(x, y) = \text{Update}_{\text{gate}}(S^*, M^*, f_{\theta^*})(x, y)$$

For a gradient-based update $\text{Update}_{\text{gate}}(S, M, f_\theta) = S - \alpha \nabla_S \hat{L}_{\text{gate}}(S; M)$, this implies:

$$\nabla_S \hat{L}_{\text{gate}}(S^*; M^*) = 0$$

i.e., S^* is a stationary point of the gatekeeper loss on the memory bank M^* .

Fixed point equation for M^* :

$$M^* = M^* \cup \{(x, y, S^*(x, y), S^*(x, y)) : (x, y) \in q, S^*(x, y) > \delta_{\text{store}}\}$$

This implies:

$$\{(x, y) \in \text{supp}(\mathcal{D}) : S^*(x, y) > \delta_{\text{store}}\} \subseteq M^*$$

i.e., M^* contains all samples that the fixed-point gatekeeper judges reliable. Since memory is monotone (B1), M^* is the limit of the growing sequence.

Fixed point equation for θ^* :

$$\theta^* = \arg \min_{\theta \in \Theta} \frac{1}{|M^*|} \sum_{(x_i, y_i, \cdot, \cdot) \in M^*} \ell_{\text{nep}}(f_{\theta}(x_i), y_i)$$

i.e., θ^* minimizes the NEP loss on M^* .

5.3 Trivial Fixed Points

Proposition 8 (Trivial Fixed Points). The following are fixed points of Φ :

1. **Zero gatekeeper:** $S_0 \equiv 0$, $M_0 = \emptyset$, θ_0 arbitrary. The system never stores any samples and never updates. (Unstable — any positive S triggers storage.)
2. **Perfect gatekeeper:** $S^*(x, y) = \mathbf{1}\{y = f^*(x)\}$, $M^* = \{(x, f^*(x), 1, 1) : x \in \mathcal{X}_0\}$, $f_{\theta^*} = f^*$. All samples are judged correctly; nothing changes. (Asymptotically stable if the NEP converges.)

3. **Constant gatekeeper:** $S(x, y) \equiv c$ for some $c \neq \delta_{\text{store}}$. If $c < \delta_{\text{store}}$, no samples are stored (like 1). If $c > \delta_{\text{store}}$, all samples are stored, M grows to the entire dataset.

Proof sketch. Each satisfies the fixed point equations by inspection. For stability claims, see Section 8. $\square$

6. Existence Conditions for Fixed Points

6.1 General Existence

Theorem 4 (Existence of Fixed Points). Suppose: 1. Φ is continuous on $\mathcal{Z}$ in the product metric $d_{\mathcal{Z}}$. 2. $\mathcal{Z}$ is compact in $d_{\mathcal{Z}}$. 3. Φ maps $\mathcal{Z}$ into itself.

Then Φ has at least one fixed point in $\mathcal{Z}$.

Proof. Apply the Brouwer fixed-point theorem (for compact convex subsets of a Banach space) or Schauder-Tychonoff (for locally convex topological vector spaces). The product structure $\mathcal{Z} = \mathcal{F} \times \mathcal{M} \times \Theta$ with $\mathcal{F}$ compact (Proposition 5) and Θ compact (under boundedness), and $\mathcal{M}$ finite, satisfies the hypotheses. $\square$

Remark. Condition 1 (continuity of Φ) is non-trivial: the argmin in Φ_{θ} is set-valued if the minimum is not unique. Formalizing Φ as a selection from a correspondence (upper hemicontinuous) allows using the Kakutani fixed-point theorem instead.

6.2 Specific Existence for SCX

Theorem 5 (Fixed Point Existence for SCX with Convex Loss).

Suppose: 1. The gatekeeper loss $L_{\text{gate}}(S; M)$ is strictly convex in S for any M . 2. The NEP loss $L_{\text{nep}}(\theta; M)$ is strictly convex in θ for any M . 3. The memory bank growth saturates: $\lim_{t \rightarrow \infty} \mathbb{P}(S_t(x, y) > \delta_{\text{store}}) = 0$ or 1 for each (x, y) .

Then there exists a unique fixed point $z^* = (S^*, M^*, \theta^*)$.

Proof sketch. Convexity ensures unique minimizers for S^* and θ^* , making Φ a function (not correspondence). Under the saturation condition, M^* is the limit of the monotone sequence. The contraction mapping principle (if Φ is a contraction) or Schauder fixed point gives existence. $\square$

*Status: **Rigorous** under the stated convexity assumptions. In practice, neural network losses are non-convex, so the conditions are violated. The theorem provides an idealized benchmark.*

6.3 Necessary Condition for Non-Trivial Fixed Point

Proposition 9 (Necessary Condition). If z^* is a fixed point with $M^* \neq \emptyset$, then the NEP student f_{θ^*} must satisfy:

$$\mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell_{\text{nep}}(f_{\theta^*}(x), y) \mid S^*(x, y) > \delta_{\text{store}}] \leq \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell_{\text{nep}}(f_{\theta^*}(x), y) \mid S^*(x, y) \leq \delta_{\text{store}}]$$

i.e., the NEP performs no worse on samples the gatekeeper admits than on samples it rejects.

Proof. At a fixed point, the gatekeeper’s storage policy and the NEP’s parameters are consistent. If the inequality were reversed, the gatekeeper would be systematically mislabeling high-loss samples as reliable, contradicting the stationarity of S^* (since S^* minimizes L_{gate} on M^* , and M^* contains only admitted samples). $\square$

7. Lyapunov Function — Explicit Candidate (DEFECT-03/04 Fix)

Status change (2026-06-28). The original Section 7 defined a vague Lyapunov candidate $V(z_t) = \mathbb{E}[L_{\text{gate}}] + \lambda \cdot \mathbb{E}[L_{\text{nep}}]$ evaluated on the acceptance-biased distribution. This was found to be **underspecified** (DEFECT-03) and to contain a **selection bias confound** (DEFECT-04): a decrease in V could reflect distribution shift rather than genuine model improvement. The following explicit candidate on a **fixed reference set** is adopted as a concrete target for future proof. The convergence claim is downgraded from “Proven” to **Conjecture**.

7.1 Concrete Lyapunov Function Candidate

Definition 23 (Concrete Lyapunov Function Candidate). Define $\Psi : \mathcal{F} \times \Theta \rightarrow \mathbb{R}_{\geq 0}$:

$$\Psi(S_t, \theta_t) = \frac{1}{|M_0|} \sum_{x \in M_0} (S_t(x, y(x)) - \hat{C}(x))^2 + \lambda \cdot \frac{1}{|V_0|} \sum_{(x, y) \in V_0} \ell(f_{\theta_t}(x), y),$$

where: - $M_0 \subset \mathcal{X} \times \mathcal{Y}$ is a **fixed reference set** (e.g., initial samples before self-evolution begins, or an external validation set). M_0 does **not** grow with t . - $V_0 \subseteq M_0$ is the subset of M_0 with externally verified clean labels, - $\hat{C}(x) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{\ell(f_m(x), y) > \tau\}$ is the consensus score computed from the M expert models (fixed), - $\lambda > 0$ is a balancing hyperparameter.

Notation note (B3 fix): The Lyapunov function is denoted Ψ (Psi), NOT Φ (Phi). The symbol Φ is reserved for the **Update Operator** $\Phi : \mathcal{Z} \rightarrow \mathcal{Z}$ defined in Document 01, Section 9. This resolves the Φ - Ψ notation collision identified in the cross-file consistency audit.

Why a fixed reference set? The original Lyapunov candidate evaluated both gatekeeper and student losses on the acceptance-biased distribution P_{S_t} . This created a **selection bias confound** (DEFECT-04/13): the gatekeeper can decrease its apparent loss by becoming more selective and admitting only “easy” samples, without improving its true discriminative ability. By evaluating on a fixed reference set M_0 that does not change with t , we ensure that a decrease in Φ reflects genuine improvement.

7.2 Component Decomposition

The Lyapunov function decomposes into two interpretable terms:

Gatekeeper term (Ψ_{gate}). Measures the squared discrepancy between the gatekeeper’s score $S_t(x, y(x))$ and the fixed expert consensus $\hat{C}(x)$ on the reference set. This term decreases when the gatekeeper’s judgments align better with expert consensus:

$$\Psi_{\text{gate}}(S_t) = \frac{1}{|M_0|} \sum_{x \in M_0} (S_t(x, y(x)) - \hat{C}(x))^2.$$

Student term (Ψ_{student}). Measures the NEP student's prediction error on the verified-clean subset V_0 . This term decreases when the student learns better representations:

$$\Psi_{\text{student}}(\theta_t) = \frac{1}{|V_0|} \sum_{(x,y) \in V_0} \ell(f_{\theta_t}(x), y).$$

7.3 Properties

Proposition 10 (Properties of Ψ). 1. **Non-negativity:** $\Psi(S, \theta) \geq 0$ for all (S, θ) , with equality iff $S_t(x, y(x)) = \hat{C}(x)$ for all $x \in M_0$ and $f_{\theta_t}(x) = y$ for all $(x, y) \in V_0$. 2. **Boundedness:** Under Assumption A3 (bounded loss $\ell \leq B$), $\Psi(S, \theta) \leq 1 + \lambda B < \infty$ since $S_t \in [0, 1]$ and $\hat{C} \in [0, 1]$. 3. **Continuity:** Ψ is continuous in (S, θ) under the product metric $d_{\mathcal{Z}}$ if ℓ is continuous and M_0 is finite.

Proof. (1) Sums of squares and non-negative losses. (2) $|S_t - \hat{C}| \leq 1$, so the squared term ≤ 1 ; $\ell \leq B$ by A3. (3) Composition of continuous functions; for finite M_0 , the empirical average is continuous in each argument. $\square$

7.4 Descent Analysis — Honest Assessment

Definition 24 (Lyapunov Descent Property). Ψ is a Lyapunov function for the dynamical system $(\mathcal{Z}, \Phi)$ (where Φ is the Update Operator) if:

$$\mathbb{E}[\Psi(S_{t+1}, \theta_{t+1}) \mid \mathcal{F}_t] \leq \Psi(S_t, \theta_t) - \eta_t, \quad \eta_t \geq 0,$$

with $\eta_t > 0$ except at fixed points.

Theorem 6 (Lyapunov Descent — CONDITIONAL, CONJECTURED). Under the following jointly sufficient conditions, the candidate Ψ satisfies the descent property:

1. **Two-timescale separation (C6’):** $\beta_t = o(\alpha_t)$ — gatekeeper updates are asymptotically slower than student updates.
2. **Student gradient descent (C2, C4):** Standard SGD with Robbins-Monro rates on α_t .
3. **Gatekeeper moves toward consensus (C3, C7):** SCXUpdate reduces the discrepancy $|S_t - \hat{C}|$ when applied to new data.
4. **Random exploration (C8):** A fraction $\varepsilon > 0$ of accepted samples are chosen uniformly at random regardless of S_t score, ensuring the reference distribution remains represented.
5. **Annealing acceptance threshold (C9):** $\gamma_t \rightarrow 0.5$ as $t \rightarrow \infty$, starting from a permissive initial value $\gamma_0 < 0.5$.

Proof sketch (gaps acknowledged). Decompose the one-step change:

$$\Psi(S_{t+1}, \theta_{t+1}) - \Psi(S_t, \theta_t) = \underbrace{[\Psi_{\text{gate}}(S_{t+1}) - \Psi_{\text{gate}}(S_t)]}_{(A)} + \lambda \underbrace{[\Psi_{\text{student}}(\theta_{t+1}) - \Psi_{\text{student}}(\theta_t)]}_{(B)}.$$

Term (A) — Gatekeeper update on reference set. The gatekeeper update $S_{t+1} = \Pi_{[0,1]}[S_t + \beta_t(\text{SCXUpdate} - S_t)]$ moves S_t toward $\hat{C}$. However, this movement is **not guaranteed** to reduce Ψ_{gate} on M_0 because SCXUpdate uses data from M_{t+1} (acceptance-biased), not from M_0 . **Gap: The alignment between SCXUpdate’s direction on M_{t+1} and the gradient of Ψ_{gate} on M_0 is not established.** Under exploration (C8), M_{t+1} eventually contains representatives from all regions, so for large t , the empirical distributions $P_{M_{t+1}}$ and P_{M_0} become close. But the rate of this con-

vergence depends on the exploration fraction ε and the data distribution's support.

Term (B) — Student update on reference set. The student is trained by SGD on M_t (acceptance-biased), not on V_0 . A standard SGD step reduces the loss on the training distribution P_{S_t} , but may **increase** the loss on the reference distribution P_{V_0} . **Gap: The relationship between $\mathbb{E}_{P_{S_t}}[\ell]$ and $\mathbb{E}_{P_{V_0}}[\ell]$ is not controlled.** Under two-timescale separation (C6'), the student takes many steps between gatekeeper updates, converging approximately to a local minimum of the current P_{S_t} . The quality of this minimum on P_{V_0} depends on the KL divergence between the distributions:

$$\mathbb{E}_{P_{V_0}}[\ell(f_\theta)] \leq \mathbb{E}_{P_{S_t}}[\ell(f_\theta)] + D_{KL}(P_{V_0} \| P_{S_t})/C,$$

which is bounded only if the gatekeeper does not systematically exclude regions of V_0 .

Overall assessment. The descent inequality $\mathbb{E}[\Psi_{t+1} \mid \mathcal{F}_t] \leq \Psi_t - \eta_t$ has **not been rigorously proven**. The decomposition above identifies two critical gaps: (i) alignment of gatekeeper update direction on biased vs. reference data, and (ii) generalization of student improvement from training distribution to reference distribution. The additional conditions (C6', C8, C9) are designed to close these gaps, but their formal sufficiency has not been established. **The descent property is therefore a conjecture, not a theorem.**

7.5 Downgraded Convergence Claim

Conjecture SE-1 (formerly “Theorem SE-1”). With the explicit Lyapunov candidate Ψ defined above and under conditions (C1')-(C9) (see Document 06 for the full condition set), the SCX self-evolution sequence satisfies:

$$\Psi(S_t, \theta_t) \xrightarrow{a.s.} \Psi_\infty \geq 0,$$

$$\sum_{t=1}^{\infty} (\alpha_t \|\nabla_{\theta} L_t(\theta_t)\|^2 + \beta_t \|\Delta S_t\|^2) < \infty \quad \text{a.s.},$$

and any limit point $(S_\infty, \theta_\infty)$ satisfies the fixed-point equations. The convergence is in the **Cesàro mean** sense rather than monotonic: the Lyapunov function may increase at individual steps but its long-run average decreases.

Honest status (2026-06-28). This is labeled a **Conjecture** because the Lyapunov descent property (Definition 24) has not been rigorously proven. The key difficulties are the selection bias cycle (DEFECT-13) and the tension between gatekeeper exploration and student stabilization (DEFECT-14). The conditions (C6'), (C8), and (C9) are proposed mechanisms to address these difficulties, but their formal sufficiency awaits proof. Numerical evidence is pending.

8. Monotonicity Analysis

8.1 Expected Monotonicity

Corollary 1 (Expected Monotonicity). Under the conditions of Theorem 6, even if the losses are not convex but the gradient updates are unbiased estimators of the true gradient:

$$\mathbb{E}[\Delta V(z_t)] \leq 0 \quad \forall t$$

where the expectation is over the random sampling of batches.

Proof. The gradient descent update ensures descent in expectation for sufficiently small step sizes (standard result for SGD with unbiased gradients). $\square$

8.2 Strict Monotonicity Gap

Proposition 11 (Strict Decrease Condition). The inequality $\Delta V(z_t) < 0$ is strict unless both of the following hold: 1. S_t is already optimal given M_{t+1} : $\nabla_S \hat{L}_{\text{gate}}(S_t; M_{t+1}) = 0$. 2. θ_t is already optimal given M_{t+1} : $\nabla_{\theta} \hat{L}_{\text{nep}}(f_{\theta_t}; M_{t+1}) = 0$.

If either gradient is non-zero, the gradient step strictly decreases the respective loss (for sufficiently small step sizes), giving $\Delta V(z_t) < 0$.

8.3 Monotonicity of Memory Bank

A separate monotonicity holds for the memory bank:

Proposition 12 (Memory Monotonicity). The sequence $\{N_t\}_{t=0}^{\infty}$ of memory bank sizes is non-decreasing:

$$N_0 \leq N_1 \leq N_2 \leq \dots$$

and converges to $N_{\infty} \leq \infty$.

Proof. $M_t \subseteq M_{t+1}$ (Assumption B1) directly implies $N_t \leq N_{t+1}$. Monotone bounded sequences converge (possibly to ∞). $\square$

8.4 Relationship Between V and N_t

Proposition 13 (Consistency of Monotonicities). A decreasing Lyapunov function $V(z_t)$ and an increasing memory size N_t are not contradictory: V measures average loss (which decreases as the system learns), while N_t measures total accumulated data (which increases). The system can simultaneously get better (lower V) and accumulate more data (higher N_t).

9. Phase Portrait

9.1 Qualitative Description

Based on the analysis above, the SCX self-evolution system has the following qualitative phase portrait:

Regime I: Initialization ($t = 0$). - S_0 derived from static SCX consensus score - M_0 contains initial reliable samples (or empty) - θ_0 from pretrained model (or random) - $V(z_0)$ is finite and bounded by Theorem 1's guarantee

Regime II: Rapid Improvement ($0 < t < T_1$). - The gatekeeper rapidly improves as it receives feedback from the growing memory bank - $V(z_t)$ decreases quickly - The NEP student begins to provide useful delayed feedback - N_t grows rapidly

Regime III: Diminishing Returns ($T_1 \leq t < T_2$). - The gatekeeper approaches its optimal performance given the available data - $V(z_t)$ decreases slowly - Memory bank growth slows (only hard or ambiguous samples remain) - The system enters a "refinement" phase

Regime IV: Saturation ($t \geq T_2$). - z_t approaches a fixed point z^* - $V(z_t)$ plateaus - N_t saturates (or grows only from genuinely novel samples) - The system has converged

9.2 Parameter Dependence

The qualitative behavior depends on several key parameters:

Parameter	Effect on Dynamics
η (noise rate)	Higher η slows convergence (more noise to filter)
μ_s (clean error)	Higher μ_s reduces initial S_0 quality, slowing early improvement
M (number of experts)	Larger M improves S_0 via tighter concentration (Theorem 1)
δ_{store}	Higher threshold: slower memory growth, higher precision
λ (Lyapunov weight)	Higher λ prioritizes NEP improvement over gatekeeper
α, β (step sizes)	Larger steps: faster convergence but risk of overshoot

9.3 Phase Diagram

We can partition the parameter space into regimes:

Region A (Convergent) $\subset \mathcal{Z}$: Initial states from which the system converges to a fixed point. Under Assumptions B1-B3, this region contains all physically realizable initial states.

Region B (Limit Cycle) $\subset \mathcal{Z}$: Initial states leading to periodic orbits. **Conjectured empty** for SCX (Conjecture 2).

Region C (Divergent) $\subset \mathcal{Z}$: Initial states from which the system diverges (e.g., $N_t \rightarrow \infty$ without convergence of S_t). This can occur if δ_{store} is too low, causing the system to accumulate noisy samples indefinitely.

Proposition 14 (Divergence Condition). If $\delta_{\text{store}} < \eta$ (the storage threshold is below the noise rate), the system may diverge in Region C: the gatekeeper stores a significant proportion of noisy samples, degrading the NEP student, which further degrades the gatekeeper, creating a vicious cycle.

Proof sketch. If $\delta_{\text{store}} < \eta$, then even a random gatekeeper would store $> \eta$ fraction of noisy samples. The stored noise contaminates M_{t+1} , which trains θ_{t+1} , potentially amplifying the error. Formalizing this requires analyzing the noise propagation eigenvalue (see Proposition 15). $\square$

10. Connection to Existing Theorems 1-3

10.1 Theorem 1: Noise Detection Guarantee

Theorem 1 provides the **initialization guarantee** for the dynamical system:

$$V(z_0) \leq 1 - \text{F1}_0 \leq \frac{1}{\eta} \sum_{s \in \mathcal{S}} \rho_s \cdot \exp(-2M\Delta_s^2)$$

This bounds the initial Lyapunov value $V(z_0)$ in terms of the noise detection F1 score.

Dynamical implication: The system starts with $V(z_0) \leq \varepsilon_0(M)$ where $\varepsilon_0(M)$ decreases exponentially in M . A larger M gives a better initialization, reducing the number of evolution rounds needed to reach a given performance level.

10.2 Theorem 2: Weak Feature Failure Lower Bound

Theorem 2 constrains the **improvement rate** of the dynamical system:

If ϕ is δ -weak, then for any t :

$$F1_t \leq F1_{\text{base}} + C_F \sqrt{\frac{\delta}{2}} + (\text{evolution improvement})$$

The “evolution improvement” term captures what self-evolution adds beyond the static baseline. **Conjectured behavior:** this improvement term itself is bounded by $O(\sqrt{\delta})$ if the state space is discovered from ϕ , because the NEP student cannot extract state information that isn’t in the features.

Corollary 2 (Improvement Bound Under Weak Features). If the feature mapping ϕ is δ -weak and the state discovery uses ϕ , then the improvement from self-evolution is bounded by:

$$\limsup_{t \rightarrow \infty} (F1_t - F1_0) \leq C \cdot \sqrt{\frac{\delta}{2}}$$

for some constant $C > 0$ depending on η and the Lipschitz constant of the F1 score.

*Status: **Conjecture** — a rigorous proof requires showing that self-evolution cannot amplify information not present in the features. The information*

bottleneck principle suggests this is true.

10.3 Theorem 3: Noise-Difficulty Unidentifiability

Theorem 3 establishes a fundamental **obstacle** to convergence to the perfect fixed point:

If the data generating process satisfies the unidentifiability conditions, then even the limiting gatekeeper S_∞ cannot achieve perfect distinction between noise and difficulty.

Proposition 15 (Residual Error at Fixed Point). Under the conditions of Theorem 3, any fixed point z^* of the SCX self-evolution system must satisfy:

$$\mathbb{E}_{x \in s_1} [|S^*(x, y_{\text{obs}}) - \mathbf{1}\{y_{\text{obs}} = f^*(x)\}|] \geq \frac{\eta\rho}{2}$$

where s_1 is the ambiguous state from the Theorem 3 construction.

Proof. Theorem 3 shows that in the ambiguous subset, any algorithm has error rate $\geq \eta\rho/2$. Since S^* induces a noise detection algorithm via thresholding, the same lower bound applies. $\square$

Corollary 3 (Non-Vanishing Lyapunov Function). Under the conditions of Theorem 3, the Lyapunov function cannot converge to zero:

$$\liminf_{t \rightarrow \infty} V(z_t) \geq c(\eta, \rho) > 0$$

where $c(\eta, \rho) = \eta \cdot \rho \cdot (\log(1/\eta) + \log(2/\rho))/2$ (approximately).

*Status: **Conjecture** — the exact constant $c(\eta, \rho)$ depends on the loss function and the data distribution. The existence of a positive lower bound follows from Proposition 15, but the precise value is not yet derived.*

10.4 Integration

The three theorems together paint a complete picture of the dynamical system:

1. **Theorem 1** (initialization): The system starts close to the optimal fixed point when M is large.
2. **Theorem 2** (speed limit): Weak features constrain how much self-evolution can improve over the initialization.
3. **Theorem 3** (fundamental barrier): Even with infinite evolution, perfect noise detection is impossible.

The self-evolution dynamics bridge these three fixed points: initialization $\rightarrow$ improvement (bounded by feature quality) $\rightarrow$ saturation (bounded by unidentifiability).

11. Summary of Proven vs. Conjectured Claims

Claim	Status	Evidence
Proposition 3 (causal structure)	Proven	By inspection of update definitions
Proposition 4 (completeness)	Proven	Standard topology results

Claim	Status	Evidence
Proposition 5 (compactness)	Proven	Arzela-Ascoli under Lipschitz bound
Proposition 6 (bounded orbits)	Proven	From B1-B3
Proposition 7 (fixed points as attractors)	Proven	By Lyapunov's theorem
Proposition 8 (trivial fixed points)	Proven	By verification
Theorem 4 (general existence)	Proven	Brouwer/Schauder fixed point theorem
Theorem 5 (unique existence, convex)	Proven	Contraction mapping / convex analysis
Proposition 9 (necessary condition)	Proven	By contradiction
Proposition 10 (Lyapunov properties)	Proven	Non-negativity, boundedness, continuity of the explicit Ψ are verified for finite M_0

Claim	Status	Evidence
Theorem 6 (Lyapunov decrease)	CONJECTURED (was “ Proven ”)	See DEFECT-03/04. The original proof assumed memory bank growth always improves losses and gradient steps always descend — this ignores the selection bias cycle and distribution shift. The corrected analysis identifies two gaps: (i) alignment of gatekeeper update direction on biased vs. reference data, and (ii) generalization of student improvement across distributions. Descent is conjectured under additional conditions (C6’, C8, C9).
Corollary 1 (expected monotonicity)	CONJECTURED (depends on Theorem 6)	Conditional on the descent property being proven
Proposition 11 (strict decrease)	CONJECTURED (depends on Theorem 6)	Conditional on Theorem 6 and stationarity conditions
Proposition 12 (memory monotonicity)	Proven	From B1

Claim	Status	Evidence
Proposition 13 (consistency)	Proven	By definitions
Proposition 14 (divergence condition)	Rigorous sketch	Requires formalizing noise propagation
Proposition 15 (residual error)	Proven	From Theorem 3
Conjecture 2 (no periodic orbits)	Conjecture	Requires extending Lyapunov argument to non-convex
Conjecture 3 (unique attractor)	Conjecture	Function-space uniqueness not guaranteed
Corollary 2 (improvement bound)	Conjecture	Information bottleneck extension of Theorem 2
Corollary 3 (non-vanishing V)	Conjecture	Exact constant not yet derived

References

The dynamical systems concepts used here are standard. Key references for the mathematical framework:

1. Strogatz, S. H. (2018). *Nonlinear Dynamics and Chaos* (2nd ed.). CRC Press. — Phase portrait analysis, Lyapunov functions.

2. Hale, J. K. (1988). *Asymptotic Behavior of Dissipative Systems*. AMS. — ω -limit sets, attractors.
3. Khalil, H. K. (2002). *Nonlinear Systems* (3rd ed.). Prentice Hall. — Lyapunov stability, invariance principles.
4. Conley, C. (1978). *Isolated Invariant Sets and the Morse Index*. CBMS Regional Conference Series. — Lyapunov functions for discrete systems.
5. Benaïm, M. (1999). “Dynamics of stochastic approximation algorithms.” *Le Seminaire de Probabilites*, 1709, 1-68. — Stochastic approximation and ODE methods for discrete dynamics.
6. Polyak, B. T., & Juditsky, A. B. (1992). “Acceleration of stochastic approximation by averaging.” *SIAM Journal on Control and Optimization*, 30(4), 838-855. — Convergence rates for iterative methods.

End of Document 02: SCX Self-Evolution as a Discrete Dynamical System

Next: Document 03 analyzes the online learning regret of the gatekeeper under delayed feedback from the NEP student.

Changelog

Date	Defect	Change	Severity
2026-06-28	DEFECT-03	Explicit Lyapunov function defined. Replaced the vague composite loss $V(z_t) = \mathbb{E}[L_{\text{gate}}] + \lambda \mathbb{E}[L_{\text{nep}}]$ with a concrete candidate $\Psi(S_t, \theta_t) = \frac{1}{ M_0 } \sum_{x \in M_0} (S_t - \hat{C})^2 + \frac{\lambda}{ V_0 } \sum_{y \in V_0} \ell(f_{\theta_t}, y)$ evaluated on a fixed reference set M_0. The function is non-negative, bounded, and well-defined for finite M_0.	FATAL

Date	Defect	Change	Severity
2026-06-28	DEFECT-04	Honest CONJECTURE status for descent property. The Lyapunov descent proof (Theorem 6) is acknowledged as not proven. Two critical gaps identified: (i) alignment of gatekeeper update direction on biased vs. reference data, and (ii) generalization of student improvement across distributions. Theorem 6 is now labeled CONDITIONAL, CONJECTURED. Corollary 1 and Proposition 11 downgraded to CONJECTURED (depend on Theorem 6).	FATAL

Date	Defect	Change	Severity
2026-06-28	DEFECT-03/13	Selection bias confound acknowledged. Section 7.4 explains why the original Lyapunov candidate (evaluated on acceptance-biased distribution) could decrease without genuine improvement. The reference-set fix (evaluating on M_0, V_0) eliminates this confound. Additional conditions (C6': two-timescale, C8: exploration, C9: annealing threshold) proposed as mitigation.	FATAL/MAJOR

Date	Defect	Change	Severity
2026-06-28	DEFECT-13/14	Cross-reference to selection bias analysis and two-timescale scheduling in Document 06 (06_fixed_point_convergence.md Sections 5.1 and 3.3).	MAJOR